



Risky decision-making in individuals with substance use disorder: A meta-analysis and meta-regression review

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Abstract

Background This review aims to identify whether risky decision-making is increased in substance users, and the impact of substance type, polysubstance use status, abstinence period, and treatment status on risky decision-making.

Methods A literature search with no date restrictions was conducted to identify case-control studies or cross-sectional studies that used behavioral tasks to measure risky decision-making in substance users. A random-effects model was performed. GRADE criteria was used to assess the quality of evidence.

Results 52 studies were enrolled. The result showed that the difference in risky decision-making performance between user groups and control groups was significant (SMD = −0.590; 95%CI = −0.849 to −0.330; $p < 0.001$; $I^2 = 93.4\%$; $P_{\text{heterogeneity}} < 0.001$). Subgroup analysis showed that users in the subgroups of alcohol ($p < 0.001$), tobacco ($p < 0.01$), cocaine ($p < 0.001$), opioid ($p < 0.001$), mixed group ($p < 0.01$), adult users ($p < 0.001$), small sample size ($p < 0.001$), large sample size ($p < 0.01$), low education ($p < 0.001$), high education ($p < 0.001$), short-abstinence period ($p < 0.001$), long-abstinence period ($p < 0.001$), without current polysubstance dependence ($p < 0.001$), and with treatment ($p < 0.001$) had increased risky decision-making when compared to the controls. On the other hand, elderly substance users with short-abstinence period showed increased risky decision-making. Moreover, current treatment status and polysubstance use may not influence the level of decision-making in substance users.

Conclusions The results show that substance use is associated with impaired risky decision-making, indicating that interventions targeting risky decision-making in substance users should be developed for relapse prevention and rehabilitation.

Keywords Substance use · Addiction · Risky decision-making · Meta-analysis · Meta-regression

Introduction

Substance use is a serious public health problem that has profound negative effects on individuals and society. It impairs social functioning, increases rate of mortality, and increases crime rate (UNODC 2018). Numerous studies have explored

the impact of impaired decision-making on the development and progression of substance dependence (Krmopotich et al. 2015; Gonzalez et al. 2015; Koffarnus and Kaplan 2018; Bickel et al. 2017; Brière et al. 2019). Sound decision-making usually refers to the ability to avoid making decisions with high-risk adverse consequences or only short-term or small-value gains. Many factors are involved in the final decision-making, including rational thinking, emotions, working memory, and declarative memory (Bechara 2000; Shohamy and Turk-Browne 2013). Among the types of decision-making, risky decision-making is a type of uncertain decision-making, that is, the probability of different outcomes of decision-making is known (Levy et al. 2010). Risky decision-making is strongly related to the compulsive/uncontrolled substance use behavior, which triggers repeated substance seeking and use behavioral pattern (Yucel and Lubman 2007). Losing control over substance use is also an important criterion for the DSM-5 diagnosis of substance use

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disorder (APA 2013). Some researchers suggested that the difference in the neural basis of risk may underlie substance dependence (Fishbein et al. 2005; Rogers et al. 1999). Therefore, it is important to understand risky decision-making of substance use.

Several studies have examined risky decision-making in substance users but results from such studies are conflicting (Murphy et al. 2012; Brand et al. 2008). The factors such as age, level of education, and type of substance abuse and so on may contribute to the heterogeneity of the results (Naude et al. 2015; Brorson et al. 2013). Abstinence period is among these factors. A study showed that patients of opioid abuse who had abstained for two weeks had a significantly increased risky decision-making compared with healthy controls (Baldacchino et al. 2015), whereas another study showed that patients with a six-week abstinence period exhibited comparable risky decision-making with healthy controls (Tolomeo et al. 2016). Substance abusers exhibited differences in neural activation during risky decision-making in the prefrontal cortex, anterior cingulate cortex, striatum, and insula when compared with healthy controls. In risk-taking tasks, substance abusers have shown greater activation in ventromedial prefrontal cortex, anterior cingulate cortex, and dorsal striatum but decreased activation in dorsolateral prefrontal and cortex ventral striatum. In addition, anterior insula activation had an inverted-U shape, such that activation was greatest when risk was highest and lowest when risk was lowest (Gowin et al. 2013). Imaging studies have shown that when the abstinence period is extended, the gray matter volume of the prefrontal cortex in substance users is increased, and cognitive function is improved (Zou et al. 2018; Parvaz et al. 2017; Petit et al. 2017; Schmidt et al. 2017). Although the neural mechanisms regulating risky decision-making by substance abusers are not completely clear, the above information suggests that the length of abstinence period will determine the individual's risky decision-making function, and will influence his future risk of relapse. Recently, a meta-analysis of opioid dependence found no association between the length of abstinence period and the impairment of decision-making (Biernacki et al. 2016). But whether this association remains significant in risky decision-making (part of decision-making) and whether it applies to abuse of different substance types requires further confirmation. In this meta-analysis, we explored whether the length of abstinence period affects the abnormality of risky decision-making of substance users through subgroup analysis.

Polysubstance use and treatment status are two other characteristics of substance users, which may affect the degree of risky decision-making. Polysubstance use problem is widespread among substance users (Connor et al. 2014), and thus increases the risk of adverse substance-related events such as relapse and mortality (Connor et al. 2014). Previous studies have also shown that polysubstance use aggravates the

impairment of cognitive functions (Stevens et al. 2007; Hoshi et al. 2007; Colzato et al. 2009). Although some studies with small sample size have shown that polysubstance use is associated with increased risky decision-making (Morgan et al. 2006; Koester et al. 2013), the extent to which polysubstance use characteristics increase risky decision-making remains unclear. Besides, therapies used in therapeutic institutions/centers are mostly approved by governmental agencies. Therefore, although it is difficult to determine the influence of each treatment on risky decision-making from each study, it is important to investigate whether such treatments improve risky decision-making. However, previous research results are inconsistent (Bieke et al. 2013, Cordovil et al. 2010, Passetti et al. 2011). Clinically, individuals whose abstinence period less than one month are often treated with detoxification, while those whose abstinence period more than a month are usually given maintenance therapy after detoxification. Therefore, optimizing the treatments for each period may help to understand the relationship between treatment of a distinct stage and risky decision-making. As far as we know, this relationship is unclear. In this study, we examined the effects of treatments on substance users who had an abstinence time of less than or more than one month.

Given the findings from previous studies, we hypothesized that substance users had increased risky decision-making. Therefore, this meta-analysis investigated three questions: (1) whether substance users exhibited increased risky decision-making compared to non-users/limited use controls and what factors may influence their risky decision-making; (2) identify the difference in magnitude of risky decision-making deficits (relative to controls) in current users and abstinent users (bounded by one month); (3) identify the impact of treatment status and polysubstance use status on the risky decision-making of current users and abstinent users (bounded by one month).

Methods

Design and registration

We conducted the review according to the registered protocol PROSPERO CRD42019118911 (<https://www.crd.york.ac.uk/PROSPERO/>) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.

Inclusion and exclusion criteria

The inclusion criteria in this meta-analysis were as follows: (1) participants: patients with substance dependence disorders or substance abusers or chronic heavy substance users; (2) intervention: behavioral tasks measuring risky decision-

making tasks assessed under conditions of ambiguity where outcomes are unknown or cannot be predicted including the Balloon Analogue Risk Task (BART) (Lejuez et al. 2002), the Cambridge Gambling Task (CGT) (Rogers et al. 1999), the Iowa Gambling Task (IGT) (Figner et al. 2009), and the Game of Dice Task (GDT) (Brand et al. 2008); (3) comparators: a non-using or limited use group; (4) outcomes: the mean effect measures (adjusted number of pumps) for BART, risk-taking for CGT, the average number of cards chosen for CGT, risky choice for GDT, and net scores for IGT; (5) type of studies: designed as a case-control study or cross-sectional study and presented in English or Chinese.

The exclusion criteria were as follows: (1) participants with other severe neurological or mental disorders listed in the Diagnostic and Statistical Manual of Mental Disorders Fifth Edition (DSM-5), Axis I; (2) conducted in animals; (3) duplication of previous publication(s). Enrollment of studies was performed by two independent investigators, and any disputes were resolved by discussion.

Search strategies

A literature search with no date restrictions was conducted on PubMed, EMBASE, Web of Science, Clinical trial, Google Scholar, and Chinese database Sinomed (<http://www.sinomed.ac.cn/>). The following keywords were used: “alcohol”, or “nicotine”, or “cocaine”, or “opioids”, or “heroin”, or “cannabis”, or “methamphetamine”, or “ecstasy”, or “MDMA”, or “GHB”, or “LSD”, or “steroids”, or “street drugs”, or “illicit” and “risky decision-making”, or “decision-making”, or “cognitive function”, or “executive function”. Final literature searches were performed in May 2019. The retrieved studies were reviewed, and duplicates were eliminated. Then, inclusion and exclusion criteria were applied to select suitable records. Finally, the titles, abstracts, keywords, and whole texts of retrieved studies were checked to exclude irrelevant ones. Also, the reference lists of the retrieved studies and recent reviews were checked manually to identify additional studies meeting the inclusion criteria.

Data extraction

The following data were extracted from eligible studies in this meta-analysis: first author’s name, publication year, sample size, proportion of female participants, mean age, level of education, type of substance use, length of abstinence, polysubstance dependence status, current treatment status, mean effect measures, and their standard deviation. Two independent reviewers double-extracted the data and cross-checked the results for discrepancy. Any deviations were resolved by discussion. A third independent reviewer assessed the coding for accuracy by randomly selecting and recoding

five articles and examining potential outliers in the data. In case of missing data, authors were contacted by e-mail. In cases where we could not obtain a feedback, studies with missing information were excluded.

Statistical analysis

Due to the anticipated heterogeneity, we used the random-effects model to calculate the overall effect size for different decision-making tasks. For each measure, we calculated Cohen’s *d* in line with the general systematic approach (Lipsey and Wilson 2000) and the associated website (http://www.campbellcollaboration.org/resources/effect_size_input.php), using means and standard deviations or standard errors where possible; occasionally, *F*, *t*, or *p* values were used with sample size to estimate the effect size. To correct for overestimation of the effect size associated with small sample sizes, the Hedge’s correction was performed to each effect size, and the inverse variance weights for each study were calculated using the corrected effect size.

After an overall effect size was computed, Hedges’ *g* was calculated as the effect size of each subgroup factor. Studies were grouped by substance type (alcohol, marijuana, tobacco, cocaine, opioids, amphetamine-like stimulants including amphetamine, methamphetamine, and MDMA and mixed drugs), sample size (small < 40, large ≥ 40), age (youth ≤ 24 years old, adult > 24 years old) (WHO 2017), level of education (low education: below high school, high education: high school completed) (Mirza and Mirza 2008; Piko and Kovacs 2010), length of abstinence (short-abstinence period ≤ one month, long-abstinence period > one month) (Schuster et al. 2016), polysubstance dependence status (dependent, intermittent use), and current treatment status (yes, no). In this study, substance users were coded as intermittent use and polysubstance dependence, according to the method described by Biernacki (Biernacki et al. 2016). Alcohol and tobacco are the most widely used legal drugs in the world, and other illegal substances are not often concurrently used by tobacco and alcohol users (Morandi et al. 2016, Kohut 2017). In illicit drug users, however, concurrent use of other substances was popular (Connor et al. 2014). Therefore, for studies on alcohol or/and nicotine that lacked criteria of polysubstance dependence status, the participants were coded by the conservative approach (i.e., coded as intermittent polysubstance use). For studies on illicit drug that lacked criteria of polysubstance dependence status, the participants were coded by the positive approach (i.e., coded as dependence for polysubstance use). For treatment status, participants recruited from clinical centers after detoxification or those who had been through a period of abstinence or participants in a stable situation were coded as substance users with current treatment.

Planned meta-regression analyses were used to examine whether any between-study heterogeneity could be explained

by the factors suggested by previous analyses which can influence the effect size e.g., substance type, sample size, age, educational level, length of abstinence, polysubstance dependence status, and current treatment status.

Heterogeneity was tested using the I^2 statistic and Q test. We considered statistical heterogeneity to be low for $I^2 \leq 40\%$, moderate for $I^2 = 30\text{--}60\%$, substantial for $I^2 = 50\text{--}90\%$, and considerable for $I^2 = 75\text{--}100\%$. Sensitivity and meta-regression analyses were used to explore the potential sources of heterogeneity. Publication bias was checked using the funnel plot and Egger's tests. The critical evaluation of the bias risk among the included studies was conducted by two independent reviewers using the Newcastle-Ottawa Scale (NOS) (Wells et al. 2014).

All statistical analyses were conducted using Stata 14.0.

Assessment of evidence in cumulative evidence

The quality of evidence for each outcome across studies was evaluated using four levels of confidence (high, moderate, low, or very low confidence) according to the Grading of Recommendations Assessment, Development and Evaluation (GRADE) criteria (Atkins et al. 2004).

Results

Characteristics of included studies

A total of 52 studies which met the inclusion criteria were enrolled, of which 28 studies used IGT, two studies used CGT, three studies used GDT, and 19 studies used BART (one study included both GDT and BART). A summary of the screening process is presented in Fig. 1. A total of 4359 participants (2173 users and 2186 controls) were included. The number of studies, control, and user participants differed between substances. 16 studies were performed on heavy drinking including 425 heavy drinkers and 404 controls, 7 studies reported on marijuana including 257 users and 252 controls, five studies reported on tobacco including 125 users and 137 controls, six studies reported on cocaine including 171 users and 185 controls, seven studies reported on opioids including 398 users and 381 controls, four studies reported on amphetamine-like stimulants including 106 users and 353 controls, and 14 studies reported on mixed substances including 568 users and 474 controls. A total of 37 studies reported 1327 users with abstinence period within one month and 16 studies reported 846 users with abstinence period over one month. In the subgroup of polysubstance dependence status, 32 studies reported 1123 users without current dependence on other substances and 23 studies reported 1050 users with current dependence on other substances. Furthermore, 30 studies reported 1211 users with treatment and 24 studies

reported 962 users without treatment. Other demographic and clinical information for each study is presented in Table 1.

In addition, the methodological quality of the studies assessed by NOS is presented in the [Supplementary materials](#). Most of the included studies were scored with six or more stars, hence were rated as high quality.

Overall effect size and heterogeneity analysis

Due to moderate heterogeneity, we used the random-effects model to synthesize the pooled effect size. The overall effect size indicated that the magnitude of the difference in risky decision-making performance between user groups and control groups was significant (SMD = -0.590 ; 95%CI = -0.849 to -0.330 ; $p < 0.001$; $I^2 = 93.4\%$; $P_{\text{heterogeneity}} < 0.001$) (see Fig. 2).

To determine the source of heterogeneity, meta-regression analyses were conducted and the results are presented in Table 2. The results revealed that there was no significant association between effect sizes and each factor except for the age subgroup.

In the subgroup analysis, risky decision-making in the subgroup of alcohol ($p < 0.001$), tobacco ($p = 0.002$), cocaine ($p < 0.001$), opioid ($p < 0.001$), mixed group ($p = 0.003$), adult users ($p < 0.001$), small sample size ($p < 0.001$), large sample size ($p = 0.009$), low education ($p < 0.001$), high education ($p < 0.001$), short-abstinence period ($p < 0.001$), long-abstinence period ($p < 0.001$), without current polysubstance dependence ($p < 0.001$), and with treatment ($p < 0.001$) was significantly different from the controls.

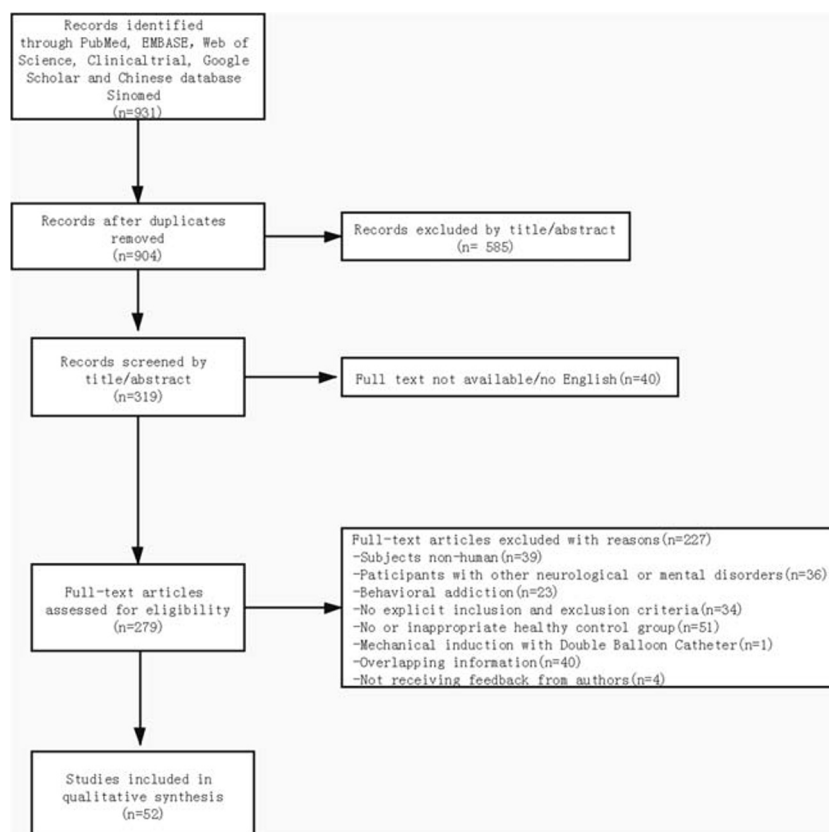
To assess the influence of outliers, the standardized residual was examined in all studies. We conducted a sensitivity analysis by removing one study at a time to evaluate the weights of individual studies on the pooled SMDs. Figure 3 shows that sensitivity analysis was relatively robust for the meta-analysis, with no reverse outcomes.

Funnel plots and Egger's test were used to reveal possible publication bias. The results showed that there was a possible overestimation of effect size (Egger's test: $P = 0.004$). Funnel plots are shown in Fig. 4.

Despite the high methodological quality, direct evidence, and precision, the studies included were characterized by moderate heterogeneity. So the existing evidence to understand risky decision-making in substance use disorder was of low quality (GRADE rating system) (see [Supplementary materials](#)).

Influence of abstinence period on risky decision-making

To explore the influence of abstinence period on risky decision-making, we analyzed the subgroups of short-abstinence period and long-abstinence period. Compared to long-

Fig. 1 The flow diagram of included studies

abstinence period, short-abstinence period as a factor would increase risky decision-making in alcohol ($SMD = -1.146$; $95\%CI = (-1.980, -0.312)$; $p = 0.007$), cocaine ($SMD = -1.914$; $95\%CI = (-3.508, -0.320)$; $p = 0.019$) as well as adult users ($SMD = -1.266$; $95\%CI = (-1.799, -0.733)$; $p < 0.001$). It seemed that there was no association between abstinence period and level of education though both low education ($SMD = -0.664$; $95\%CI = (-1.162, -0.166)$; $p = 0.009$) and high education ($SMD = -1.214$; $95\%CI = (-1.892, -0.535)$; $p < 0.001$) users showed significantly increased risky decision-making in short-abstinence period. Furthermore, short-abstinence period would increase risky decision-making in users with intermittent polysubstance use ($SMD = -1.353$; $95\%CI = (-1.945, -0.761)$; $p < 0.001$). Short-abstinence period would increase risky decision-making whether users receiving treatment ($SMD = -0.983$; $95\%CI = (-1.564, -0.401)$; $p = 0.001$) or not ($SMD = -0.707$; $95\%CI = (-1.181, -0.233)$; $p = 0.003$).

Influence of current polysubstance dependence status

Further subgroup analysis was performed for current polysubstance dependence status. Intermittent polysubstance use would increase risky decision-making in alcohol ($SMD = -1.148$; $95\%CI = (-1.744, -0.552)$; $p < 0.001$) and adult

($SMD = -1.204$; $95\%CI = (-1.678, -0.730)$; $p < 0.001$) users. There was no association between current polysubstance dependence status and level of education though both low education ($SMD = -0.546$; $95\%CI = (-0.980, -0.111)$; $p = 0.014$) and high education ($SMD = -1.757$; $95\%CI = (-2.500, -1.013)$; $p < 0.001$) users with intermittent polysubstance dependence showed significantly increased risky decision-making. Intermittent polysubstance use would increase risky decision-making whether users receiving treatment ($SMD = -0.969$; $95\%CI = (-1.457, -0.480)$; $p < 0.001$) or not ($SMD = -1.023$; $95\%CI = (-1.625, -0.421)$; $p = 0.001$).

Discussion

The primary purpose of this meta-analysis was to systematically identify risky decision-making of substance users, as well as the influence of substance use-related factors and demographic factors on risky decision-making. In this study, we identified 52 studies comparing risky decision-making of substance users and controls. Results revealed significantly increased risky decision-making in substance users compared to controls. These results are consistent with the previous meta-analysis about opioid users (Biernacki et al. 2016; Baldacchino et al. 2012). In addition, elderly substance users with short-abstinence period who were dependent on one

Table 1 Demographic and clinical information for each study

Study	Behavioral measure	Substance users			Substance use	Abstinence	Polysubstance use status	Treatment status	Control group		
		N	Mean age	Level of education					N	Mean age	Mean education year or range
Crowley et al. (2006)	BART	20 (3)	16.1	NA	Mixed	NA	Dependence	Y	20 (5)	16.5	NA
Gonzalez et al. (2007)	IGT	17 (5)	33.8	12.4 yrs.	Alcohol	Range from 15 to 30 days	Dependence	Y	19 (7)	31.1	14.9 yrs.
Gonzalez et al. (2007)	IGT	16 (4)	35.4	11.7 yrs.	Mixed	> 1 month	Dependence	Y	19 (7)	31.1	14.9 yrs.
Noël et al. (2007)	IGT	30 (10)	45.8	10.7 yrs.	Alcohol	Range from 15 to 30 days	Intermittent use	Y	30 (11)	44.1	10.8 yrs.
Tanabe et al. (2007)	IGT	20 (10)	35	11.0 yrs.	Mixed	> 1 month	Dependence	Y	16 (11)	37	14.0 yrs.
Verdejo-Garcia et al. (2007)	IGT	12 (0)	36.8	13.3 yrs.	Cocaine	< 7 days	Dependence	Y	14 (0)	30.9	13.6 yrs.
Verdejo-Garcia et al. (2007)	IGT	11 (0)	25.7	13.0 yrs.	Marijuana	< 7 days	Dependence	Y	14 (0)	30.9	13.6 yrs.
Brand et al. (2008)	GDT	18 (4)	31.11	9.33 yrs.	Opoids	Mean 14.38 (7.41) days	Intermittent use	Y	18 (5)	31.28	10 yrs.
Dolan et al. (2008)	IGT	18 (11)	33.7	11.9 yrs.	Mixed	Range from 15 to 30 days	Dependence	Y	18 (8)	27.7	16.6 yrs.
Miranda et al. (2009)	IGT	22 (0)	29.3	12.9 yrs.	Alcohol	> 1 month	Intermittent use	N	21 (0)	22.6	14.5 yrs.
Cordovil et al. (2010)	IGT	35 (18)	48.4	NA	Alcohol	< 7 days	Intermittent use	Y	22 (8)	44.4	NA
Hammers et al. (2010)	BART	23 (14)	19.1	12.5 yrs.	Mixed	< 1 month, subjects used drugs within the past month	Dependence	N	23 (12)	19.1	12.4 yrs.
Van der Plas et al. (2009)	IGT	33 (18)	37.9	13.8 yrs.	Alcohol	Range from 15 to 30 days	Intermittent use	Y	36 (17)	28.9	16.2 yrs.
Van der Plas et al. (2009)	IGT	27 (14)	34.5	11.4 yrs.	Cocaine	Range from 15 to 30 days	Intermittent use	Y	36 (17)	28.9	16.2 yrs.
Van der Plas et al. (2009)	IGT	38 (25)	30.7	11.7 yrs.	Mixed	Range from 15 to 30 days	Intermittent use	Y	36 (17)	28.9	16.2 yrs.
van Toor et al. (2011)	IGT	31 (14)	36.4	NA	Mixed	> 1 month	Dependence	Y	31 (15)	36.3	NA
Dean et al. (2011)	BART	26 (15)	20.2	13.8 yrs.	Tobacco	< 1 day, all smoking participants reported smoking daily and had urine cotinine levels $\geq$ 100 ng/mL and expired carbon monoxide (CO) levels $\geq$ 8 ppm	Intermittent use	N	38 (20)	19.8	14 yrs.
Kim et al. (2011)	IGT	23 (0)	32.7	11.3 yrs.	Alcohol	Range from 15 to 30 days	Intermittent use	Y	21 (0)	30.5	15.1 yrs.

Table 1 (continued)

Study	Behavioral measure	Substance users			Substance use	Abstinence	Polysubstance use status	Treatment status	Control group		
		N	Mean age	Level of education					N	Mean age	Mean education year or range
Kim et al. (2011)	GDT	23 (0)	32.65	11.26 yrs.	Alcohol	At least 2 weeks	Intermittent use	Y	21 (0)	30.52	15.14 yrs.
Schepis et al. (2011)	BART	12 (2)	17.2	<24 yrs., high school	Tobacco	< 1 day, smokers were eligible if they smoked ≥ 5 cigarettes daily and had urine cotinine levels ≥ 350 ng/ml	Intermittent use	N	15 (7)	15.7	<24 yrs., high school
Wesley et al. (2011)	IGT	16 (7)	26.4	NA	Marijuana	< 7 days	Intermittent use	N	16 (10)	26.6	NA
Cousijn et al. (2013)	IGT	32 (11)	21.4	NA	Marijuana	< 7 days	Intermittent use	N	41 (15)	22.2	NA
Gonzalez et al. (2012)	BART	65 (0)	20.8	13.5 yrs.	Marijuana	< 1 month, subjects used 3 (1.25) days ago	Dependence	N	65 (14)	20.3	13.6 yrs.
Thompson et al. (2012)	BART	30 (10)	32.5	11.5 yrs.	Mixed	1.34 (0.98) yrs.	Dependence	Y	28 (15)	37.3	13.2 yrs.
Cavalea et al. (2013)	BART	22 (11)	16.2	<24 yrs., adolescents	Tobacco	< 1 month, subjects reported using 5 or more cigarettes per day over the past 30 days	Intermittent use	N	17 (8)	16.2	<24 yrs., adolescents
Bieke et al. (2013)	IGT	37 (8)	31.6	NA	Mixed	Range from 15 to 30 days	Dependence	Y	31 (4)	28.1	NA
Galván et al. (2013)	BART	18 (9)	19.47	13.73 yrs.	Tobacco	< 1 week, carbon dioxide concentration ≥ 6 ppm in breath and/or urinary cotinine ≥ 200 ng/ml was required for inclusion in the smoker group	Dependence	N	25 (11)	19.08	13.56 yrs.
Korreich et al. (2013)	IGT	25 (10)	45.7	NA	Alcohol	Range from 15 to 30 days	Intermittent use	Y	25 (6)	39.0	NA
Korreich et al. (2013)	IGT	25 (5)	39.5	NA	Mixed	Range from 15 to 30 days	Intermittent use	Y	25 (6)	39.0	NA
Li et al. (2013)	IGT	124 (0)	29.3	NA	Opioid	> 1 month	Intermittent use	Y	43 (0)	26.3	NA
Mellentin et al. (2013)	IGT	30 (10)	34.2	NA	Mixed	Range from 15 to 30 days	Dependence	Y	17 (4)	36.7	NA
Verdejo-Garcia et al. (2013)	IGT	86 (25)	21.5	NA	Marijuana	< 7 days	Intermittent use	N	58 (19)	23.2	NA
Xiao et al. (2013)	IGT	14 (6)	17.3	NA	Alcohol	< 7 days	Intermittent use	N	14 (9)	17.1	NA
Zorlu et al. (2013)	IGT	17 (0)	47	7.6 yrs.	Alcohol	Range from 15 to 30 days	Intermittent use	N	16 (0)	46.7	8.0 yrs.
Baloni et al. (2014)	IGT	34 (14)	54.8	13.2 yrs.	Cocaine	< 7 days	Intermittent use	N	39 (17)	56.4	14.0 yrs.
Baldacchino et al. (2015)	CGT	24 (0)	26.3	10.8 yrs.	Opioids	< 3 days, subjects used heroin 29.5 (2.7) days in the last 30 days	Intermittent use	Y	28 (0)	24.12	15.4 yrs.

Table 1 (continued)

Study	Behavioral measure	Substance users			Substance use	Abstinence	Polysubstance use status	Treatment status	Control group		
		N	Mean age	Level of education					N	Mean age	Mean education year or range
Brevers et al. (2014)	IGT	30 (8)	44.5	14.1 yrs.	Alcohol	Range from 15 to 30 days	Intermittent use	Y	30 (6)	41.5	13.6 yrs.
Gorini et al. (2014)	GDT, BART	18 (8)	38.4	14.42 yrs.	Cocaine	At least 2 weeks before the test	Intermittent use	Y	18 (8)	36.8	15.56 yrs.
Hanson et al. (2014)	BART	24 (7)	19.5	NA	Marijuana	> 1 m. subjects used 52.9 (52.7) days ago	Intermittent use	N	34 (10)	18.9	NA
Kohno et al. (2014)	BART	26 (13)	35.68	13.0 yrs.	Methamphetamine	Range from 4 to 7.62 days	Dependence	N	27 (11)	33.88	13.62 yrs.
Le Berre et al. (2014)	IGT	30 (4)	43.7	10.5 yrs.	Alcohol	Range from 7 to 14 days	Intermittent use	Y	45 (7)	44.8	10.7 yrs.
Yan et al. (2014)	IGT	58 (0)	35.7	8.4 yrs.	Opioid	> 1 month	Intermittent	Y	60 (0)	34.3	9.5 yrs.
Zorlu et al. (2014)	IGT	12 (0)	53.9	7.6 yrs.	Alcohol	> 1 month	Intermittent	Y	13 (0)	52.1	8.0 yrs.
Briggs et al. (2015)	IGT	24 (12)	NA	NA	Tobacco	< 7 days	Intermittent use	N	21 (12)	NA	NA
Briggs et al. (2015)	IGT	23 (11)	NA	NA	Tobacco	< 7 days	Intermittent use	N	21 (12)	NA	NA
Fischer et al. (2015)	BART	23 (7)	27.35	12.18 yrs.	Marijuana	< 7 days, 22 (95.7%) subjects were tested positive for urine	Dependence	N	24 (12)	29.21	13.48 yrs.
Krmpotich et al. (2015)	BART	70 (32)	34.3	12.2 yrs.	Mixed	1.2 (0.9) yrs.	Dependence	Y	92 (37)	33.4	14.1 yrs.
Ahn et al. (2016)	BART	44 (11)	29.32	12.61 yrs.	Opioids	1.74 (2.35) yrs.	Intermittent use	N	81 (22)	24.27	13.8 yrs.
Ahn et al. (2016)	BART	39 (13)	23.44	13.1 yrs.	Amphetamine	0.40 (0.60) yrs.	Intermittent use	N	81 (22)	24.27	13.8 yrs.
Ahn et al. (2016)	BART	58 (9)	25.95	12.83 yrs.	Mixed	0.77 (0.96) yrs.	Dependence	N	81 (22)	24.27	13.8 yrs.
Romero-Ayuso et al. (2016)	BART	25 (8)	27	< 12 yrs.	Cocaine	> 3 months	Dependence	Y	25 (15)	30	< 12.0 yrs.
Malloir-Bague et al. (2016)	IGT	106 (32)	21.7	11.7 yrs.	Amphetamine-like stimulants	> 1 month	Dependence	Y	194 (154)	25.0	14.5 yrs.
Tolomeo et al. (2016)	CGT	25 (0)	36.6	NA	Opioids	> 6 months	Intermittent use	N	50 (0)	28	NA
Tolomeo et al. (2016)	CGT	48 (0)	30.2	NA	Opioids	> 6 weeks	Intermittent use	Y	50 (0)	28	NA
Zeng et al. (2016)	IGT	58 (25)	25.7	9.5 yrs.	Amphetamine-like stimulants	< 7 days	Dependence	Y	51 (25)	23.3	10.5 yrs.
Zeng et al. (2016)	IGT	57 (26)	42.5	9.3 yrs.	Amphetamine-like stimulants	< 7 days	Dependence	Y	51 (25)	23.3	10.5 yrs.
	BART	152 (31)	16.23	NA	Mixed		Dependence	N	37 (17)	16.05	NA

Table 1 (continued)

Study	Behavioral measure	Substance users			Substance use	Abstinence	Polysubstance use status	Treatment status	Control group		
		<i>N</i>	Mean age	Level of education					<i>N</i>	Mean age	Mean education year or range
Claus et al. (2018)						< 1 month, subjects reported using their primary used drug more than one time in the past month					
Lannoy et al. (2017)	BART	20 (8)	20.25	> 12 yrs., undergraduate	Alcohol	< 1 week, subjects were recruited from university and not asked to be abstinent before the test; 18.85 (10.06) total units per week	Intermittent use	N	20 (7)	21.2	> 12.0 yrs., undergraduate
Maurage et al. (2018)	IGT	38 (9)	47.0	12.5 yrs.	Alcohol	< 7 days	Intermittent use	Y	38 (9)	46.7	12.5 yrs.
Wang et al. (2018)	BART	56 (0)	43.36	11.13 yrs.	Alcohol	Mean 50.58 days	Intermittent use	Y	33 (0)	42.88	9.09 yrs.
Hobkirk et al. (2019)	BART	37 (12)	45.3	12.65 yrs.	Cocaine	< 1 month, subjects used 10.46 (SD = 7.57) days out of the past 30	Dependence	N	35 (13)	41.29	13.97 yrs.

Abbreviations: *BART*, Balloon Analogue Risk Task; *CGT*, Cambridge Gambling Task; *IGT*, the Iowa Gambling Task; *GDT*, the Game of Dice Task; *N*, no; *Y*, yes; *NA*, not applicable; *yrs.*, years

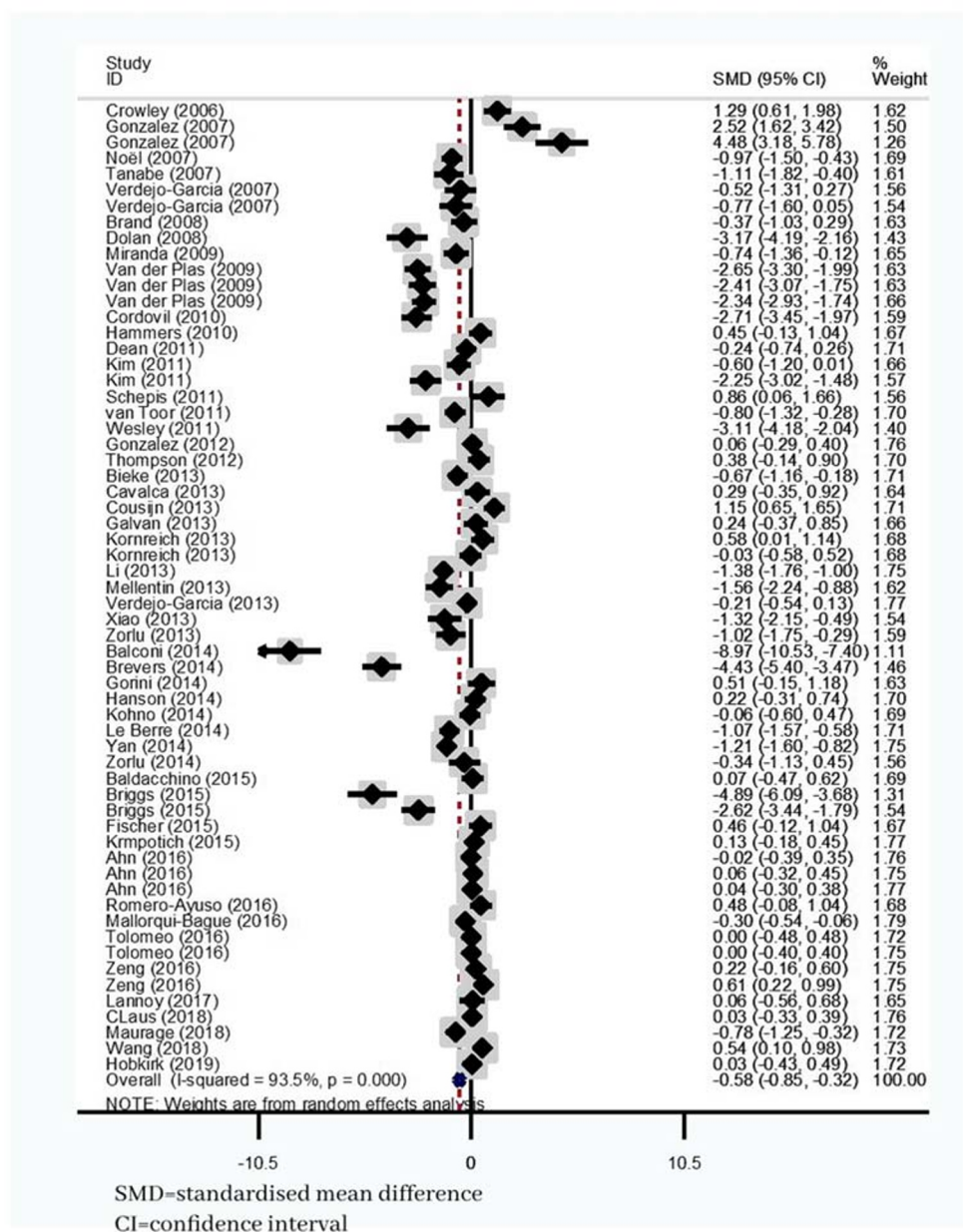


Fig. 2 Forest plot of the pooled effect size

substance showed severe impairment in decision-making. Furthermore, the relationship between treatment status and risky decision-making was found in users with short-abstinence period.

Risky decision-making has shown to correlate with substance use and likely contributed to the development, maintenance, and severity of addiction (Fishbein et al. 2005; Baldacchino et al. 2012). In this study, increased risky decision-making was observed in most substance type except for marijuana and amphetamine-like stimulants. Risky decision-making was consistent with their distinct neurological damage in the prefrontal cortex and striatum that underlie cognition and reward (Clay et al.

2008; Gozzi et al. 2011; Hu et al. 2015; Zhang et al. 2018; Burger et al. 2018). For cocaine, further subgroup analysis showed that risky decision-making occurred in users with short-abstinence period. For alcohol, this study found that the users with short-abstinence period and/or intermittent polysubstance use exhibited significant deficits. It seems that cocaine and alcohol directly result to increased risky decision-making as well as they are associated with the time for treatment. In contrast, marijuana and amphetamine-like stimulants did not influence risky decision-making. Previous studies have found some evidence for a causal relationship between abuse of these two drug abuse and cognitive decline

Table 2 Results of meta-regression analysis

Variables	Regression coefficient (SE)	95% CI	p value
Paradigm	−.423 (.248)	−.921 to .076	.095
Substance type	−.111 (.118)	−.347 to .125	.349
Abstinence	.785 (.497)	−.212 to 1.781	.120
Sample size	.055 (.518)	−.985 to 1.095	.916
Age	− 1.137 (.474)	− 2.088 to −.186	.02-0*
Level of education	−.154 (.300)	−.756 to .447	.609
Polydrug status	1.031 (.524)	−.021 to 2.083	.055
Treatment status	−.532 (.483)	− 1.501 to .436	.275

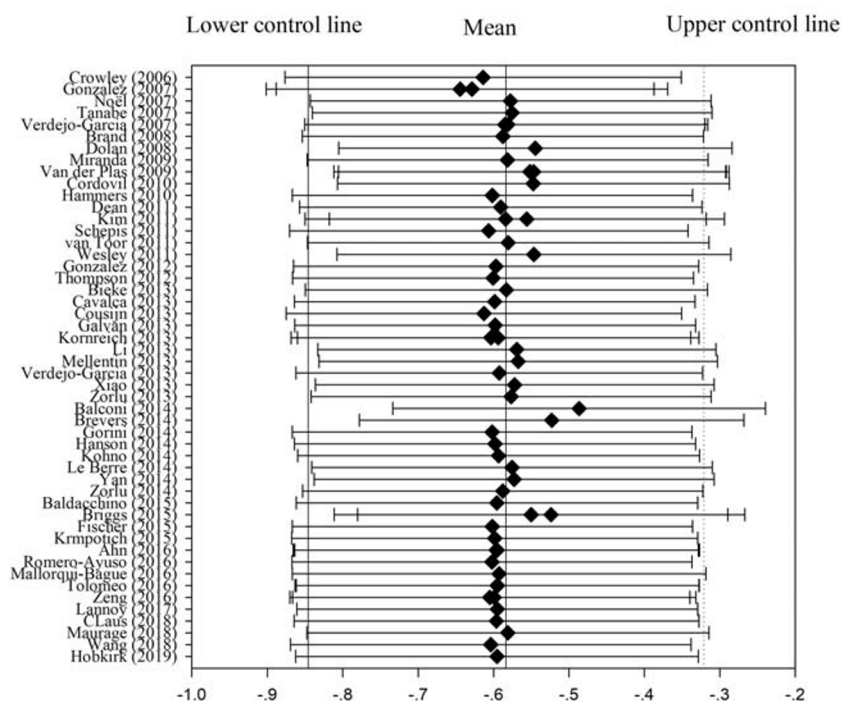
Abbreviations: *SE*, standard error; *CI*, confidence interval* <0.05

including decision-making (Droutman et al. 2019; Delibas et al. 2018). Occurrence of cognitive deficits may contribute to a high relapse rate in users of stimulants such as methamphetamine (Brecht and Herbeck 2014). Indeed, impulsive behavior is a common feature of patients using marijuana (Baldacchino et al. 2015). Although the neurotoxicity caused by the abuse of

amphetamines is well recognized, the existing behavioral studies do not provide enough evidence to explain the difference in decision-making impairment between abuser group and healthy subjects. Further studies are needed to examine the neurocognitive characteristics of amphetamine-like stimulants and marijuana during

Fig. 3 Results of “one-study removed” sensitivity analysis for all studies included

Meta-analysis estimates, given named study is omitted



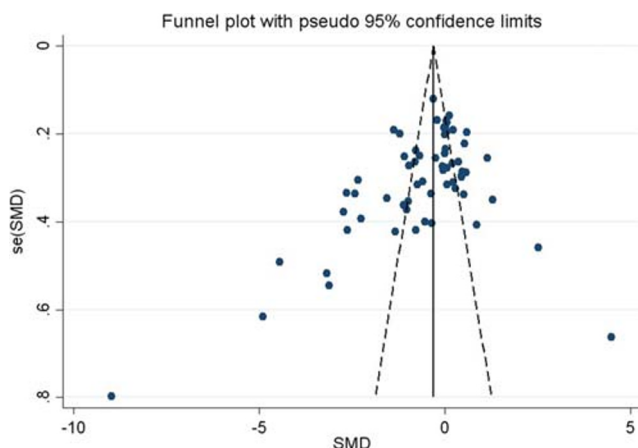


Fig. 4 Funnel plot with pseudo 95% confidence limits

decision-making and help identify intervention strategies.

Age and education are two important factors associated with substance use. It was reported that substance users with low education levels were more likely to become addicted and had worse substance use-related clinical consequences (Mehta et al. 2012; Blum et al. 2014). However, we found users with both low education and high education had increased risky decision-making, which is inconsistent with previous results. On the other hand, our results showed that adult users (> 24 years old) tended to have poorer decision-making function. Due to the lack of enough information regarding their drug-use duration, we speculated that elderly substance users may have longer drug-use time. It has also been shown that impulsivity differs with age and decision-making is thought to involve a cooperative relationship between an impulsive system activated by immediate rewards and an inhibitory control system (Argyriou et al. 2017; Verdejo-Garcia et al. 2008).

Functional magnetic resonance imaging (fMRI) research indicates that individuals with SUD show impaired decision-making associated with altered brain activation in executive control and reward processing regions (Aron and Paulus 2007; Fowler et al. 2007). Evidence that there is aberrant activity in key components of the pre-frontal cortex-limbic network has led researchers to suggest that weakened ability to accurately process information about options and control behaviors paves way for choices that offer immediate, rather than delayed, rewards (Ray and Sinha 2008). But the impaired structure and function progressively recovers as the abstinence period increases (Zou et al. 2018; Parvaz et al. 2017; Verdejo-Garcia et al. 2008), but it is not clear whether this is accompanied with improvement of behaviors. We found that the length of abstinence period may not be associated

with the degree of risky decision-making, which is consistent with findings of a previous meta-analysis including patients with opioid dependence only (Lipsey and Wilson 2000; Mintzer et al. 2005). On the other hand, these results suggest that although the risky decision-making is expected to recover as the abstinence period increases, the improvement in the first one and a half years of abstinence was minimal (Zhang et al. 2011). Our results seem to support this opinion.

To further understand the characteristics of different abstinence periods, this study also explored the impact of treatment status and polysubstance use status on risky decision-making. In general, current users usually receive less than four weeks of treatment for detoxification and withdrawal symptoms (CAODAPAT 2017; Drummond et al. 2011; David et al. 2018). After withdrawal, patients are then given rehabilitation and relapse prevention treatment. However, few studies have explored the cognitive functions such as decision-making at different treatment stages. This study shows that treatment did not significantly affect risky decision-making of abstinent users. We found that risky decision-making was worse among the current users receiving treatment. The reason may be that the clinical condition of individuals being treated is often worse, and some patients are less likely to go to the hospital (Hiller et al. 2009; Levin et al. 2006). This is not surprising, as these studies differ in treatment methods and are not clearly reported, so they could not be systematically quantified and examined in this meta-analysis. Few original studies have analyzed the impact of treatment on risky decision-making hence there is no sufficient literature for meta-analysis. Thus, our results serve as a good reference value. In addition, the results showed that polysubstance use does not influence the level of decision-making deficit in substance users, which is also consistent with previous research (Biernacki et al. 2016). This suggests that increased risky decision-making may be directly related to the primary substance use, which affects the brain function (Cheng et al. 2013; Lyoo et al. 2006).

Publication bias was reported in some studies which compared substance users and controls; however, the level of publication bias in this analysis should be explained with caution given the strict inclusion criteria and different paradigms included (Higgins and Green 2011; Verdejo-Garcia et al. 2018). In this study, a range of electronic databases were used to perform a comprehensive literature search, but it is likely that a small number of eligible unpublished or recently published literature were not included.

Although this meta-analysis provides insights into the risky decision-making in substance users, there are several limitations to this review. Firstly, this study mainly included case-control studies. Few longitudinal studies are available to analyze risky decision-making of different stages in substance use disorder. Second, our results may not be applicable to all

populations of substance users because specific conditions such as co-morbid psychiatric disorders or limited use were excluded in this study. Furthermore, we could not analyze the effect of various treatment methods on risky decision-making because of the lack of relevant data. Nevertheless, we found a significant correlation between treatment status and risky decision-making. In terms of strengths, the results of this study provide relatively reliable evidence that substance users display increased risky decision-making. Further, we show that such deficits are related to substance type, age, condition of polysubstance use, abstinence period, and treatment status. It is important to pay attention to these substance-related characteristics when formulating strategies to improve risky decision-making.

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